

AI FOR HOUSING HACKATHON

A part of AI Horizons 2026 · Pittsburgh

PARTICIPANT PACKET

Kickoff Friday, September 25 · Build September 26–27 · Fully virtual

Register by Thursday, September 24, 11:59 p.m. ET

Everything you need to know before you start building.

Start Here

You're going to spend a weekend building an AI tool for one of the least glamorous and most consequential problems in the country: why it is so hard to build housing people can afford.

You are not building a pitch deck, you are building a working prototype that someone at a planning department, a nonprofit developer, or a community organization could actually open and use. You'll have about thirty-nine hours, real public data, and access to people who do this work for a living.

The Short Version

What	A virtual hackathon where teams build AI prototypes for housing challenges, using real data and examples from Pittsburgh and Allegheny County
Kickoff	Friday, September 25 — live, with workshops. Recorded
Build window	Saturday, September 26, 9:00 a.m. ET to Sunday, September 27, 11:59 p.m. ET
Where	Entirely online. Slack for the community, Zoom for live sessions
Register by	Thursday, September 24, 11:59 p.m. ET
Who	College students, startup builders, and anyone interested in tackling housing problems. 18 and over
Teams	1 to 5 people. You sign up as an individual and form a team in Slack
Cost	Free
Prizes	The 1st, 2nd, and 3rd place college student and startup teams will share a \$30,000 cash prize, plus public recognition and visibility across the AI Horizons and partner network.

You don't need to be an expert. Plenty of people here will be writing their first AI application or working on their first housing problem. This is an opportunity to learn, grow, and potentially impact the world for the better.

Timeline

All times Eastern.

When	What	What You Should Do
Thu, Sept 17	Registration opens	Register
Sat, Sept 19	Slack community opens	Join and introduce yourself. Teams form here, and people who show up early get the best ones
Wed, Sept 23	Schedule and resources published	Read the challenge briefs and browse the data. You're encouraged to do this before the event

When	What	What You Should Do
Thu, Sept 24, 11:59 p.m.	REGISTRATION CLOSES	Last call
Fri, Sept 25	- Kickoff & Housing Subject Matter Training - Cursor Training	Show up. We cover the tracks, the rules, and judging, then run workshops on the data and tools
Sat, Sept 26, 9:00 a.m.	BUILDING BEGINS	Write your first line of code. Not before
Sat, 10:00 a.m. – 6:00 p.m.	Office hours in Slack with experts and mentors	Use them early. Ten minutes Saturday morning saves six hours Sunday
Sun, 10:00 a.m. – 6:00 p.m.	Final office hours in Slack	Come with specific questions
Sun, Sept 27, 11:59 p.m.	SUBMISSIONS CLOSE	Hard deadline. No extensions
Mon–Wed, Sept 28–30	Judging	Nothing required of you
Thu, Oct 1	Winners notified	Check your email
Fri, Oct 2	Winners announced publicly	Share your project

Who Can Participate

- You must be 18 or older.
- College students of every kind — undergraduate, graduate, community college, bootcamp, part-time, and recent graduates.
- Startup builders and early-stage teams.
- Anyone interested in solving problems related to housing.
- You don't need to be at a partner institution, in Pittsburgh, or in Pennsylvania. Participation is virtual and open.
- Organizers, mentors, subject matter experts, judges, and sponsor representatives can't compete for prizes.

Teams may mix — a student and a startup founder on the same team is fine, as long as everyone meets the criteria and everyone actually participates.

The Three Challenge Tracks

Pick one. These came from people who work on housing in this region. Full briefs, with personas and data sources, are published separately and worth reading before the weekend.

1. Development Feasibility Navigator

This challenge seeks an AI-driven tool that generates a 'Development Ease Score' for potential sites, highlighting regulatory bottlenecks, required variances, and infrastructure gaps so cities and developers can prioritize fast-track sites and identify areas needing policy or resource intervention.

Without a clear way to quantify these barriers, development pipelines remain slow, costly, and unpredictable

2. Housing Production, Rents & Household Flow Observatory

An interactive observatory combining administrative, market, and community indicators to show how the housing system changes over time.

Helps planners, policymakers, advocates, and journalists see what's actually happening before arguing about what to do. The hard part is that the data doesn't agree with itself — different geographies, time periods, and definitions of the same word.

3. Housing Typology, Equity & Climate Matchmaker

A decision-support tool matching locations with plausible housing types and surfacing tradeoffs across demand, transit, equity, and climate resilience.

Helps municipalities decide what housing fits where, and community partners who want a say in that answer. The hard part is that the tradeoffs genuinely conflict — a tool that pretends there's one right answer is less useful than one showing you what you're giving up.

These datasets describe real neighborhoods and real households. The strongest submissions are explicit about who benefits, who might be harmed, and what the tool gets wrong. "We don't have good data on X, so our tool doesn't claim to answer it" is a strength, and judges score it that way.

Teams

- You register as an individual, not as a team.
- Teams are 1 to 5 people. Going solo is allowed, but three or four is where most good projects land.
- Teams form in Slack from September 19. Bring people you know or find them there.
- Every member has to actually participate. Adding a name to a roster isn't participating.

If You Don't Have a Team

1. Post in the team-formation channel when Slack opens: what you're good at, what you want to learn, which track interests you, your time zone.
2. Reply to two other people's posts. Replying works considerably better than posting and waiting.

Four people with different skills beats five with the same one. The teams that do well usually have someone comfortable with data, someone who can build an interface, and someone who can explain why any of it matters to a housing practitioner. That third person is the one teams forget to recruit and the one judges notice is missing.

Rules

Adapted from the Major League Hacking standard hackathon rules.

Building

1. All work on your project happens during the hackathon. Building starts at 9:00 a.m. Eastern on Saturday, September 26 and stops at the submission deadline.
2. You can bring an idea. Thinking about the problem, sketching, reading the briefs, and exploring the data beforehand are encouraged.
3. You can't bring code. Reusing your own prior project code isn't allowed, and neither is open-sourcing something beforehand so you can use it during the event.
4. Your idea doesn't have to be original. If someone's built something similar, build yours anyway and build it well.
5. You can use libraries, frameworks, open-source code, public datasets, and third-party APIs. List them in your README.
6. When time is up, stop. You can fix a small bug you find while recording your demo. You can't add features.

Code

1. Your code must be in a public repository, and it has to stay public after the event to remain eligible for prizes.

2. Your commit history must be intact and start from kickoff. We check this — it's how the no-prior-code rule is enforced rather than merely stated.
3. Strip your API keys and credentials before making the repo public. Every hackathon has at least one team that doesn't.

Everything Else

1. One project per team. You can't submit the same project to another hackathon running at the same time.
2. Everyone on the team participates.
3. Ask for help. Mentors, experts, organizers, and other teams are all fair game. Helping another team is in the spirit of this, not against it.
4. Organizers can disqualify a team for breaking these rules or the code of conduct. This is rare and we'd like it to stay that way.

AI Tools

Use them. This is an AI hackathon and it would be strange to ask otherwise. Coding assistants, code generation, image generation, and large language models are all permitted.

Two conditions. Be honest about what you used. Be sure to list your AI tools in your submission and be ready to answer questions about them. And build something: a thin wrapper around an existing AI product isn't a project, and judges can tell the difference between a team that used AI to build faster and a team that asked AI to build it.

Who Owns What You Build

You do. Entering doesn't transfer any ownership to the organizers, partners, or sponsors. We ask permission to show your project publicly and that's the extent of it. If a partner wants to talk about taking the work further, that conversation happens after the announcement and it's entirely yours to accept or decline.

How to Submit

Submissions go through a Google Form. The link is published in Slack and at kickoff. Start your submission Saturday afternoon, not Sunday night — you can edit it right up to the deadline.

Item	Details
Team and member details	Team name, every member's name and email, affiliation
Challenge track	Which of the three you're entering
Project title and description	What it does, who it's for, what you'd build next
Demo video	3 to 5 minutes. See below
Public repository link	Publicly accessible, commit history intact
Data sources used	Which datasets, and where you got them
AI tool disclosure	Which AI tools you used and how
Attestation	Everyone is 18+ and no code predates kickoff

The Demo Video

The most underrated part of the submission. Judges watch it before they look at anything else, and some will form most of their opinion from it. Three to five minutes.

- Start by saying the name of the hackathon and who you are.
- Show the thing working. A screen recording of the actual tool beats slides every time.
- Be honest about what's functional and what's mocked. Judges respect "this part works, this part is a placeholder" far more than a smooth video that turns out to be a prototype of a prototype.
- Record it during the hackathon weekend, and **keep it public afterward**.

If you didn't finish, submit anyway and show what you built. An unfinished project that's honest about where it stopped scores better than you'd expect, and considerably better than not submitting.

There are no late submissions. The form closes at 11:59 p.m. Sunday and we can't reopen it for one team without being unfair to every team that made it.

Judging and Prizes

Judging is asynchronous, Monday September 28 through Wednesday September 30. Every project is seen by at least three judges, so no single person determines your outcome. The highest-scoring submissions win.

Judges come from housing practice, universities, technology, and the investment community. Some will know your track's subject matter deeply and some will be evaluating your technical work. This is why explaining your project clearly matters as much as building it well.

Evaluation Criteria

Problem Value

Does the prototype address a costly, frequent, or consequential housing bottleneck identified in the challenge briefs (e.g., development feasibility, missing-middle housing typologies, or policy/permitting navigation)?

User Fit & Usability

Is the interface intuitive, plain-language, and practical for target end-users (such as municipal planners, small developers, nonprofits, or community partners) to adopt?

Technical Execution

Does the working demo/prototype reliably perform its core functional tasks during testing and presentation?

Data & AI Integrity

Are sources, statutes, and baseline assumptions accurately cited and grounded? Are privacy constraints (no PII/sensitive data) and model uncertainties handled responsibly? Is there a clear human-in-the-loop/escalation path for consequential decisions?

Actionability

Does the output directly assist or accelerate real-world decisions in housing development, policy, planning, or permitting rather than merely presenting abstract analytics?

Continuation Potential

Is there a credible path to post-event testing, maintenance, community ownership, or pilot adoption by public/civic partners (e.g., Allegheny County, City Planning, URA, PHFA)?

MANDATORY ELIGIBILITY & COMPLIANCE CHECKS:

- **Originality:** Must be a new, ground-up build created during the hackathon window (no pre-existing products/pitches).
- **Required Deliverables:** Working code repository, demo video (3-5 min), documentation, data/source citations, and limitations statement.
- **Responsible Framing:** Tool must be positioned strictly as decision support, not binding legal, financial, or zoning advice.

Prizes

The 1st, 2nd, and 3rd place college student and startup teams will share a \$30,000 cash prize, plus public recognition and visibility across the AI Horizons and partner network.

Winners are notified Thursday, October 1 and announced publicly Friday, October 2.

Help During the Weekend

Office Hours

When	Who's There
Saturday, September 26, 10:00 a.m. – 6:00 p.m.	Housing policy, planning, permitting, and development experts, plus technical mentors across AI/ML, NLP, geospatial, and tooling
Sunday, September 27, 10:00 a.m. – 6:00 p.m.	The same, weighted toward technical triage as the deadline approaches

Use these early. The most common regret at the end of a hackathon is not asking a question on Saturday that would have redirected the whole project. Teams routinely build for twelve hours on a misunderstanding of how permitting works that a planner would have corrected in four minutes.

Mentors will explain how something works, point you at the right data, tell you what you're misunderstanding, help you debug, and tell you honestly when your scope is too big.

Where to Ask What

If you need...	Go to
A teammate	The team formation channel
Housing or policy expertise	The subject matter expert channel, during office hours
Technical help	The technical assistance channel, during office hours
Help with the submission form	The general help desk channel
Anything about rules, deadlines, or logistics	The general help desk channel
To report a code of conduct concern	See below — there's a confidential route

Code of Conduct

This event adopts the Major League Hacking Code of Conduct. The full text is linked in Slack and you agree to it when you register. The one-page version:

- Harassment isn't tolerated. Not in public channels, not in direct messages, not in video calls, not as a joke.
- This applies to everyone — participants, mentors, experts, judges, organizers, and sponsor representatives. Nobody's role exempts them.

- If something happens to you or you see something happen, tell us. You can report anonymously via Slack DM to one of the organizers.
- We take reports seriously and act on them. Consequences can include removal from the event.

Beyond the formal standard: this is a weekend where a lot of people are doing something unfamiliar in front of strangers. Be the kind of participant who makes that easier. Help the team that's stuck. Answer the beginner's question properly.

Questions People Actually Ask

I've never been to a hackathon. Is this going to be over my head?

No. A substantial share of participants are first-timers. Workshops start from the basics, mentors are there specifically for people who are stuck, and there's a channel for questions you think are too basic to ask. They aren't.

Do I have to stay up all night?

No, and we'd rather you didn't. Sleep-deprived teams make bad decisions and ship worse projects. Nothing in the judging rewards suffering.

Can I work on an idea I've already been thinking about?

Yes. Ideas are fine, encouraged even. Code isn't. Start the implementation Saturday morning.

What if my project doesn't work by Sunday?

Submit it anyway and be straight about where it stopped. That scores better than not submitting, which scores zero.

Do I need to know anything about housing?

Not on Friday. You'll need to know something by Sunday night, which is what the office hours and challenge briefs are for. Reading the briefs beforehand puts you well ahead.

Do I need to use any specific tools or platform?

No, but we are providing Cursor tokens.

What if I can only participate part of the weekend?

Participate anyway. Tell your team upfront what your availability is. Teams handle this fine when it's stated in advance and badly when it's a surprise on Sunday.

Can I keep working on this afterward?

Yes, and several teams do. It's your project. Past hackathon projects have turned into capstones, class projects, and companies.

Links and Contacts

Register	Registration Link
Slack community	https://join.slack.com/t/aihorizonshack/shared_invite/zt-48Impa6mu-vVXgw6M8ZUBMc5ocP1v3RQ
Challenge briefs	Tracks

Data resources	Data Resources
Submission form	https://docs.google.com/forms/d/e/1FAIpQLSfDK_aD-miOV3D92BI4NOFa1Skb8_u-GTCH5j1FE-VEWTr4DQ/viewform?usp=sharing&ouid=103197394082273502758
Code of conduct reports	eben@aistriketeam.com
General questions	info@aihorizonspgh.com
Event website	AI for Housing Hackathon

See you on the 25th. Come with questions.